

Exercise 1: Write a Python program to convert a distance from kilometers to miles.

Object Name:

Write a program called **Asn2_q1.py**.

Input:

The program should contain a function called **convert_to_miles** that performs the following:

1. Prompts the user to enter a distance in kilometers (distance_km).

Expected Output:

The program should:

1. Convert the input distance from kilometers to miles using the formula:
$$\text{Miles} = \text{Kilometers} \times 0.6214$$
2. Display the converted distance in miles, formatted to two decimal places.

For example:

If the user inputs distance_km = 10, the output should be:

10 kilometers is equal to 6.21 miles.

Exercise 2: Write a Python program to create a simple math quiz with addition problems.

Object Name:

Write a program called **Asn2_q2.py**.

Input:

The program should contain a function called **math_quiz** that performs the following:

1. Generates two random integers (num1 and num2) for the quiz.
2. Displays the addition problem in the format:

```
num1
+ num2
-----
```

3. Prompts the user to input their answer to the problem.

Expected Output:

The program should:

1. Check if the user's answer is correct.
2. If the answer is correct, display a congratulatory message (e.g., "Correct! Great job!").
3. If the answer is incorrect, display the correct answer in a message (e.g., "Incorrect. The correct answer is 469.").

Exercise 3: Write a Python program to guess a user's personality type based on their answers to three yes/no questions.

Object Name:

Write a program called **Asn2_q3.py**.

Input:

The program should contain a function called **who_am_i** that performs the following:

1. Asks the user three yes/no questions:
 - "Do you like being around a lot of people?"
 - "Do you enjoy trying new things?"
 - "Do you prefer to plan things out?"

Expected Output:

The program should:

1. Evaluate the user's responses (yes or no) to the three questions.
2. Based on the responses, output one of the following messages:
 - If all answers are yes, print: "You are adventurous and outgoing!"
 - If all answers are no, print: "You are thoughtful and reserved."
 - If the answers are a mix of yes and no, print: "You have a balanced personality."

For example:

If the user answers:

- Yes to "Do you like being around a lot of people?"
- No to "Do you enjoy trying new things?"
- Yes to "Do you prefer to plan things out?"

The output should be:

You have a balanced personality.

If the user answers yes to all questions, the output should be:

You are adventurous and outgoing!

Exercise 4: Write a Python program to implement the FizzBuzz game for numbers from 1 to 20.

Object Name:

Write a program called **Asn2_q4.py**.

Input:

The program should contain a function called **fizz_buzz** that performs the following:

1. Iterates through numbers from 1 to 20.

Expected Output:

The program should:

1. Print "Fizz" for numbers that are multiples of 3.
2. Print "Buzz" for numbers that are multiples of 5.
3. Print "FizzBuzz" for numbers that are multiples of both 3 and 5.
4. Print the number itself if it is not a multiple of 3 or 5.

For example:

The output should be:

```
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
16
17
Fizz
19
Buzz
```

Exercise 5: Write a Python program to find the smallest and largest numbers from user input.

Object Name:

Write a program called **Asn2_q5.py**.

Input:

The program should contain a function called **find_min_max** that performs the following:

1. Prompts the user to enter numbers one by one.
2. Stops taking input when the user enters 0.

Expected Output:

The program should:

1. Ignore the input 0 when finding the smallest and largest numbers.
2. Display the smallest and largest numbers from the entered values.
3. If no numbers are entered (other than 0), display an appropriate message (e.g., "No numbers were entered.").

For example:

If the user inputs:

Enter a number: 5

Enter a number: 2

Enter a number: 9

Enter a number: 0

The output should be:

Smallest number: 2

Largest number: 9

If the user inputs 0 as the first number, the output should be:

No numbers were entered.